Voice SQL

Ask your company data questions by voice in English, Hindi or Hinglish, and hear the answer back, with security layers that never trust the AI's SQL.

The problem

Business users often depend on analysts or SQL skills to get answers from company data, and most tools assume English typing. Voice plus an LLM can remove that barrier for India's multilingual workforce, but an LLM that writes SQL can be talked into leaking or destroying data. Enterprises will not adopt voice analytics unless it is safe by design.

The solution

Voice SQL lets a user speak or type a question (for example, a Hindi voice query about an employee's salary). The app transcribes it, converts it to a read-only SQL query, runs it under strict controls, and replies in text and in speech, in the same language as the question. Requests that try to modify data, or ask for data the user's role cannot see, are blocked, announced out loud, and logged.

How it works

Voice -> Gnani Prisma (speech to text) -> Gemini (natural language to SQL) -> **security layers** -> SQLite -> spoken summary -> Gnani Timbre (text to speech)

Gnani Prisma	Speech to text with automatic Indic language detection; handles English, Hindi and Hinglish voice queries.
Gnani Timbre	Speaks every answer, in Hindi for Hindi questions, and speaks the security alert when a request is blocked.
Google Gemini	Translates the question into SQL and writes the short spoken summary. It is pluggable: the design treats the model as untrusted.

Security by design (five layers)

- **SQL validation:** one read-only statement only; forbidden keywords, system tables and unknown tables rejected.
- **Database authorizer:** the database itself refuses anything except reading allowed tables and columns.
- **Role-based access:** a Manager role cannot read salary data; enforced inside the database, not just in the UI.
- **Resource limits:** read-only connection, 2 second query limit, 100-row cap, input and audio size limits, per-session rate limiting.
- **Safe output and audit:** user and AI text is escaped (no HTML or Markdown injection), API keys are redacted from errors, every request is logged.

Tested behaviour (shown in the demo video)

A Hindi voice question returns the correct employee and salary, answered in Hindi speech. As the Manager role, the same salary request is denied. A "Delete all employees" request is blocked by validation, with a spoken alert and an audit-log entry. Layer-level tests also confirmed the database refuses hidden-column reads, extension loading and runaway queries.

Who it helps

HR, finance and operations managers who need quick numbers; field and hands-free users; and teams more comfortable speaking Hindi or Hinglish than writing queries in English.

Honest limitations and next steps

The demo uses a small SQLite database (employees, projects). Roles are a demo selector and would come from real login (SSO) in production; rate limiting is per session and would move to a server-side gateway. Query results are sent to the LLM for the spoken summary, so sensitive deployments need a private model: the next step is to try Gnani's open-weights Evon model for that. Further planned work: testing on noisy telephonic audio and more Indian languages.

Built by Prithviraj Vijay Sharma for the Great Indian AI Internship Challenge by Gnani.ai. #GnaniAI #GreatIndianAIInternshipChallenge